Cloud Computing & DevOps – AWS Handbook

SESSION 6

AWS Elastic Beanstalk

Deploying Applications Without Managing Infrastructure

Duration	2 Hours
Prerequisites	AWS Account, EC2 and Web Hosting Knowledge
Learning Outcome	Deploy applications using Elastic Beanstalk and understand Platform as a Service (PaaS).

Why Are We Learning This?

Managing servers manually can be time-consuming. AWS Elastic Beanstalk allows developers to deploy applications quickly without worrying about provisioning servers, configuring load balancers, or managing infrastructure.

Architecture Diagram

Developer
↓
Application Code
↓
Elastic Beanstalk
↓
EC2 + Load Balancer + Auto Scaling
↓
Users

Key Concepts

PaaS	Platform as a Service
Elastic Beanstalk	AWS application deployment service
Environment	Running application instance
Application Version	Deployable package version
Auto Scaling	Automatic resource scaling
Load Balancer	Distributes incoming traffic

Practical Activities

• Create an Elastic Beanstalk application

• Create an application environment

• Upload application package

• Deploy the application

• Access the application URL

• Monitor application status

Expected Outputs

Application deployed successfully, environment health displayed as healthy, and application accessible through the generated URL.

What You Should See

Paste Screenshot Here

Elastic Beanstalk Dashboard

Paste Screenshot Here

Application Configuration

Paste Screenshot Here

Environment Creation

Paste Screenshot Here

Deployment Status

Paste Screenshot Here

Application Running in Browser

Journal Writing Template

Aim

To deploy and manage a web application using AWS Elastic Beanstalk.

Theory

Elastic Beanstalk is a Platform as a Service that automates application deployment and infrastructure management.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Application deployed successfully and environment health status remained healthy.

Conclusion

AWS Elastic Beanstalk simplified application deployment without requiring manual infrastructure management.

Viva Preparation

1. What is AWS Elastic Beanstalk?
2. What is Platform as a Service (PaaS)?
3. What are the advantages of Elastic Beanstalk?
4. What is an Environment in Elastic Beanstalk?
5. What is Auto Scaling?
6. What is a Load Balancer?
7. Difference between EC2 deployment and Elastic Beanstalk deployment?

Troubleshooting Box

Problem	Possible Cause
Deployment failed	Incorrect application package
Environment unhealthy	Configuration issue
Application not loading	Service startup failure
URL inaccessible	Environment still initializing

Industry Relevance

Elastic Beanstalk is used by organizations that want faster application deployment without managing infrastructure. It is particularly useful for development, testing, and small-to-medium production workloads.

Session Summary

Students learned how Platform as a Service simplifies application deployment by abstracting infrastructure management while still providing scalable and reliable hosting.